Sharwama: Product Requirements Document

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1. Overview

Sharwama is a web app where customers in one city browse a single shawarma shop's menu, order, pay online through Bachs, and receive the order by third-party rider delivery or pick it up. Staff run the shop from an admin dashboard: menu, orders, rider booking, promos, delivery zones and store hours.

Working name: Sharwama (replace when the brand name is final).

2. Goals and non-goals

Goals

- A customer can go from opening the site to a paid order in under 2 minutes on a phone.
- No paid order is ever lost, and no customer is charged twice for one order.
- Staff can run a full lunch rush from the admin dashboard on a phone or tablet.
- The site feels fast and appetizing on mobile data.

Non-goals for v1

- Multi-vendor marketplace or multiple branches
- Delivery outside the launch city
- Rider app, live map tracking, delivery-provider API integration (riders are booked manually)
- Loyalty points, reviews, native mobile apps, dark mode
- Scheduled orders (planned for v1.1)

3. Users

User	Description	Main needs
Customer	Orders from a phone, usually at lunch or evening. Guest checkout, optional account.	See the menu fast, order in few taps, pay safely, know when food arrives

User	Description	Main needs
Admin / staff	Shop owner or staff at the counter, on a phone, tablet or laptop	See new paid orders immediately, move orders through stages, book riders, mark items sold out
Rider	External, no login	Receives order details from staff by phone or WhatsApp

4. Scope by release

Feature	v1	v1.1	Later
Menu, options, cart, checkout	Yes		
Bachs payment, webhook, reconciliation	Yes		
Delivery with zone-based fees	Yes		
Pickup	Yes		
Promo codes	Yes		
Store hours and open/closed toggle	Yes		
Admin order board, menu manager, zones, promos	Yes		
SMS or WhatsApp order updates	Yes		
Guest checkout, optional accounts, order history	Yes		
Scheduled orders		Yes	
Refunds through API			Yes (needs Bachs access)
Delivery provider integration, live tracking			Yes
Multi-branch, loyalty, reviews			Yes

5. Functional requirements

Priority: M = must have for v1, S = should have.

5.1 Menu and browsing

ID	Requirement	Pri
FR-1	Show categories and items with photo, name, description and price in naira	M
FR-2	Sticky category tabs that scroll to sections	M
FR-3	Items can have options (size, extras, drinks) with price changes; some options required, some optional	M
FR-4	Sold-out items are visible but disabled, with a Sold out label	M
FR-5	Menu is served from cache and updates within 1 minute of an admin change	S

5.2 Cart

ID	Requirement	Pri
FR-6	Add, remove and change quantity; cart persists in the browser	M
FR-7	Sticky cart bar on mobile showing item count and subtotal; drawer on desktop	M
FR-8	Cart revalidates against current menu at checkout (price changes, sold-out items) and tells the customer what changed	M

5.3 Checkout

ID	Requirement	Pri
FR-9	Guest checkout: name, phone (required), email (optional)	M
FR-10	Choose delivery or pickup	M
FR-11	Delivery: address, landmark and area from a list of active zones; fee shown before payment	M

ID	Requirement	Pri
FR-12	Areas outside active zones show a clear not-delivering-here message	M
FR-13	Promo code field with instant validation and visible discount	M
FR-14	Server calculates every total; client prices are never trusted	M
FR-15	Minimum order amount (configurable)	S
FR-16	Order notes (for example, no onions), max 200 characters	S
FR-17	Prevent double submission with an idempotency key	M

5.4 Payment (Bachs)

ID	Requirement	Pri
FR-18	Create a Bachs checkout session per order and redirect the customer	M
FR-19	Order becomes paid only from a verified webhook, never from the redirect	M
FR-20	Webhook handler is idempotent and checks the paid amount against the order total	M
FR-21	Unpaid orders expire after 30 minutes	M
FR-22	Reconciliation job every 5 minutes checks pending orders against Bachs	M
FR-23	Late payments and amount mismatches are flagged for admin review	M

5.5 Order tracking

ID	Requirement	Pri
FR-24	Order page with status stepper, items, totals and rider name and phone once dispatched	M

ID	Requirement	Pri
FR-25	Page polls for status every 10 seconds	M
FR-26	Guest orders are reachable by an unguessable link sent by SMS or WhatsApp	M
FR-27	Customer can cancel before the order reaches preparing	S

5.6 Promo codes

ID	Requirement	Pri
FR-28	Percentage or fixed-amount codes with expiry date, max uses and minimum order	M
FR-29	One code per order; usage counted only when the order is paid	M

5.7 Notifications

ID	Requirement	Pri
FR-30	Send SMS or WhatsApp messages on: payment confirmed, out for delivery (with rider details), ready for pickup, cancelled	M
FR-31	Notification failures are logged and retried once; they never block order flow	M

5.8 Admin dashboard

ID	Requirement	Pri
FR-32	Order board: new paid orders appear within 10 seconds with a sound and a visual highlight	M
FR-33	Move orders only to valid next statuses; every change is logged with who and when	M
FR-34	Book rider step: enter rider name and phone, mark out for delivery	M

ID	Requirement	Pri
FR-35	Menu manager: categories, items, options, photos, sold-out toggle	M
FR-36	Delivery zones and fees, promo codes, store settings	M
FR-37	Flagged payments list (mismatches, late payments)	M
FR-38	Daily summary: orders, revenue, cancelled count	S

5.9 Store hours

ID	Requirement	Pri
FR-39	Weekly opening hours plus a manual Open/Closed override	M
FR-40	When closed: menu stays browsable, a banner shows opening time, checkout is disabled	M

6. Key user flows

Customer: delivery order

1. Open site, browse menu, tap an item, choose options, add to cart.
2. Open cart, review, go to checkout.
3. Enter name and phone, choose delivery, enter address and pick area. See fee and total.
4. Optionally apply a promo code.
5. Tap Pay. Redirected to Bachs checkout. Pays.
6. Returns to the site: sees Checking your payment, then Order confirmed.
7. Receives a confirmation message. Follows the order page until delivery.

Staff: handling an order

1. New order appears on the board with a sound.
2. Accept it, then mark Preparing, then Ready.
3. Delivery: book a rider outside the app, enter rider name and phone, mark Out for delivery. The customer is notified.

4. Mark Delivered when confirmed. Pickup: mark Picked up.

Failure paths to design

- Payment fails or is abandoned: order expires after 30 minutes; customer can retry from the order page while it is still pending.
- Item sells out while in the cart: warn at checkout and let the customer remove it.
- Customer pays after the order expired: admin decides to honor or refund.
- Shop closes during checkout: block payment creation and explain.

7. Order status model

Status	Meaning	Changed by
pending_payment	Order created, waiting for payment	System
paid	Payment confirmed by webhook	Webhook only
accepted	Staff acknowledged the order	Admin
preparing	Being made	Admin
ready	Ready for rider or pickup	Admin
out_for_delivery	Rider booked and dispatched	Admin
delivered / picked_up	Completed	Admin
cancelled	Cancelled before completion	Admin, or customer before preparing
expired	Never paid within 30 minutes	Scheduled job
refunded	Money returned (manual through Bachs for now)	Admin

Rules: statuses only move forward along valid transitions; every change is written to a status history table; invalid jumps are rejected by the server.

8. Payments with Bachs

- Each order gets one Bachs checkout session, created with an idempotency key based on the order ID.

- Verify the webhook signature on the raw request body using the official SDK. Prefer the V2 signature header and keep the default 5-minute timestamp tolerance.
- Mark paid only when the event is a successful collection, the reference matches, and the amount equals the order total.
- The success redirect only shows a pending state and polls the order.
- Money is stored as integer kobo. Prices and names are snapshotted on order items.

To verify in the Bachs sandbox before building

1. Can a checkout use an ad-hoc amount, or must each order be a one-off Bachs product?
2. Is NGN fully supported at checkout and settlement?
3. Fees, settlement timing, and refund access through the API.
4. Exact webhook event names and payload shape for success and failure.

9. Non-functional requirements

Area	Requirement
Availability	Reliable during lunch and evening peaks; short maintenance windows off-peak are acceptable
Consistency	Strong consistency for orders and payments; menu may be cached up to 1 minute
Performance	Largest paint under 2.5s on 4G; checkout API calls under 1s; initial JavaScript for the menu page under 200 KB
Capacity	Design for about 100 orders/day, peak about 20/hour; a single API instance and one Postgres database are enough
Durability	Daily database backups with a tested restore; no paid order lost
Security	Webhook signature check, rate limits on orders, promo validation and auth, role guards on admin routes, secrets only in environment variables, validated DTOs, CORS locked to own domains
Privacy	Collect only name, phone, email (optional) and address; publish a privacy policy covering personal data under Nigerian data protection law; delete or anonymize data on request

Area	Requirement
Observability	Error tracking, structured logs, uptime check on /health; alerts for webhook verification failures, amount mismatches and missed webhooks found by reconciliation
Accessibility	WCAG 2.1 AA

10. Data model (summary)

users, addresses, delivery_zones, menu_categories, menu_items, item_options, orders, order_items, payments, promo_codes, order_status_history, store_settings (hours, override, minimum order).

- orders: type (delivery or pickup), status, subtotal, delivery_fee, discount, total, address snapshot, promo reference, rider name and phone, placed time, public order token.
- order_items: name and unit price snapshot, quantity, selected options.
- payments: unique Bachs reference, status, amount, raw webhook payload, paid time.

11. API (summary)

Base path /api/v1.

- Public: GET /menu, GET /delivery-zones, POST /promo/validate, POST /checkout/quote, POST /orders (requires an Idempotency-Key header), GET /orders/:id, POST /orders/:id/cancel.
- Auth: register, login, refresh, GET /me.
- Webhook: POST /webhooks/bachs (raw body, signature verified, no auth guard).
- Admin: orders list and detail, PATCH status, PATCH rider, cancel, menu CRUD, availability toggle, zones CRUD, promos CRUD, flagged payments, store settings.
- Conventions: integer kobo, cursor pagination, error shape with code, message and details, status changes only through the status endpoint.

12. Architecture and stack

- Modular monolith. Frontend: Next.js and TypeScript on Vercel. Backend: NestJS and TypeScript. Database: PostgreSQL with Prisma.
- NestJS modules: auth, menu, zones, promos, orders, payments, notifications, admin, common.

- Monorepo: apps/api, apps/web, packages/shared (types and status enums).
- Background jobs through scheduled tasks, guarded by a Postgres advisory lock. Polling for status updates. No Redis or message queue in v1.
- Environments: local, staging (Bachs sandbox keys), production (live keys). Local webhook testing through a tunnel.

13. Design system

13.1 Brand feel

Warm, fast, appetizing, no-nonsense. Think toasted wrap, spice and fire: burnt orange on a cream background, with real food photography doing most of the selling. Lots of whitespace, large tap targets, one clear action per screen.

13.2 Color tokens

Token	Value	Use
primary	#C2410C (burnt orange)	Primary buttons, links, active tab, focus ring. White text on it is about 5:1 contrast
primary-hover	#9A3412	Hover and pressed state
primary-soft	#FFEDD5	Selected chips, light highlights
accent	#F59E0B (saffron)	Promo badges and small highlights only. Never for text on light backgrounds
ink	#1C1917	Headings and body text
ink-muted	#57534E	Secondary text, descriptions
bg	#FFFBF5 (cream)	Page background
surface	#FFFFFF	Cards, sheets, drawers
border	#E7E5E4	Dividers and input borders
success	#15803D	Paid, delivered
warning	#B45309	Attention states
danger	#B91C1C	Errors, cancelled
info	#1D4ED8	Informational notes

Rule of use: neutrals make up about 90% of any screen. Orange is reserved for actions and selected states, so the eye always finds the next step. No dark mode in v1.

13.3 Admin status colors

Every status chip uses a color plus an icon and a text label, never color alone.

Status	Color
pending_payment	Stone gray
paid	Blue
accepted, preparing	Amber
ready	Orange
out_for_delivery	Purple
delivered, picked_up	Green
cancelled, expired, refunded	Red or gray

13.4 Typography

- Headings: Bricolage Grotesque, weights 600 and 700.
- Body and UI: Inter, weights 400, 500 and 600. Fallback: system-ui.
- Load through next/font with swap display. Maximum two families.
- Scale (px): 12, 14, 16, 18, 20, 24, 30, 36, 48. Body text is at least 16px on mobile so iOS does not zoom into inputs.
- Line height 1.5 for body, 1.2 for headings.
- Prices use tabular numbers and the format ₦2,500: naira symbol, thousands separator, no decimals.

13.5 Spacing, layout and alignment

- 4px base grid. Spacing steps: 4, 8, 12, 16, 24, 32, 48, 64.
- Mobile first. Breakpoints: 640, 768, 1024, 1280.
- Container max width 1200px. Side padding 16px mobile, 24px tablet, 32px desktop.
- Menu layout: mobile is a single-column list of horizontal cards (text on the left, 96px photo on the right, price and add button below the text); tablet is 2 columns; desktop is 3 columns of vertical cards with the photo on top.

- Text is left-aligned by default. Center only the hero headline, empty states and order-confirmed screens.
- Prices and totals are right-aligned in order summaries and admin tables.
- Forms: single column, labels above fields, inline error text under the field.
- Buttons: full width on mobile inside forms and sticky bars; auto width on desktop. The primary action sits at the bottom on mobile and bottom-right on desktop.
- Sticky elements on mobile: category tabs at the top and the cart bar at the bottom. Add bottom padding so content never hides behind the cart bar.
- Corner radius: 12px cards and sheets, 10px buttons and inputs, full pill for chips.
- Shadows: subtle by default (0 1px 2px at 6% ink); elevated for drawers and sheets (0 4px 12px at 8% ink).
- Touch targets at least 44px; primary buttons 52px tall.

13.6 Components

Button (primary, secondary, ghost, danger), MenuCard, CategoryTabs, OptionSheet, QuantityStepper, CartBar, CartDrawer, AddressForm, PromoInput, OrderSummary, StatusStepper, StatusChip, Toast, Skeleton, Banner (store closed, info), EmptyState.

UI stack: Tailwind CSS, shadcn/ui built on Radix primitives (dialogs, tabs, focus handling), lucide icons at 20 or 24px with a 1.75 stroke.

13.7 Animation

Principles: motion confirms an action or explains a change, nothing more. Animate only transform and opacity. UI feedback stays at or under 300ms. Animation never blocks input. Respect prefers-reduced-motion by switching to instant changes or simple fades.

Interaction	Animation	Duration
Button press	Scale to 0.97 and back	100ms, ease-out
Card hover (desktop)	Lift 2px and deepen shadow	150ms
Add to cart	Add button becomes a quantity stepper; cart bar slides up from the bottom; count badge bumps from 1.15 to 1	200 to 250ms
Option sheet and cart drawer	Slide up on mobile, slide in from the right on desktop; backdrop fades	250ms open, 200ms close
Category tab change	Active underline slides to the new tab	200ms
Menu loading	Skeleton shimmer loop	1.2s loop

Interaction	Animation	Duration
Page content	Simple fade in; no page-transition effects	150ms
Order status stepper	Progress line fills to the new step; the current step has a soft pulse	300ms fill, 2s pulse loop
Toasts	Slide down and fade in; auto-dismiss	200ms in, dismiss after 3s
Hero	One fade-up on first load only	300ms
Admin new order	Row highlight fades from soft orange to normal; short sound	2s fade

Not allowed: parallax, autoplay video, scroll-jacking, looping decorative animation, and any animation on payment fields other than validation feedback.

Implementation: CSS transitions first. Use Framer Motion only for the drawer, sheet and list enter or exit transitions.

13.8 Imagery and icons

- Real food photos, 4:3 crop, warm natural light, consistent background. Served through next/image as WebP, card images under about 80 KB, with a blurred placeholder.
- Items without a photo show a simple illustrated placeholder, never a broken image.
- One hero image, lazy-load everything below the fold.

13.9 Accessibility

- WCAG 2.1 AA contrast, visible focus ring (2px primary with 2px offset), full keyboard support, focus trapping in dialogs.
- Alt text on food images, labels on every input, errors announced to screen readers.
- Status is always conveyed by text and icon as well as color.

13.10 Voice and microcopy

Short, friendly, direct. No slang overload, no jargon.

Moment	Copy
Add button	Add to cart
Checkout button	Pay ₦6,500

Moment	Copy
Store closed banner	We are closed right now. We open at 10:00 AM.
Out of area	We do not deliver to this area yet. You can choose pickup instead.
Payment pending	Checking your payment. This usually takes a few seconds.
Order confirmed	Order confirmed. We are on it.
Sold out	Sold out
Expired order	This order expired because payment was not completed. Start a new order.

14. Screens

Customer: Home and menu, Item option sheet, Cart, Checkout, Payment return (checking), Order tracking, Order history (logged-in), Login and register.

Admin: Login, Order board (columns by status on desktop, tabs on mobile), Order detail, Menu manager, Delivery zones, Promo codes, Flagged payments, Store settings.

15. Notification messages

Event	Message
Payment confirmed	Order #1042 confirmed. Total ₦6,500. We will update you when it is ready. Track: link
Out for delivery	Order #1042 is on the way. Rider: Musa, 0803 000 0000.
Ready for pickup	Order #1042 is ready for pickup.
Cancelled	Order #1042 was cancelled. If you paid, we will refund you. Contact us on the shop number.

16. Success metrics

- Orders per day and week
- Checkout conversion (checkout started to paid)

- Payment success rate and share of orders recovered by reconciliation
- Median time from paid to delivered
- Share of orders needing manual intervention (flagged payments, cancellations)
- Page load time on mobile 4G

17. Risks and open questions

Item	Detail	Owner
Bachs fit	Ad-hoc amounts, NGN support, fees, settlement, refund access (section 8)	Dev, before building payments
Launch city and zones	Which areas and what fee per zone	Shop owner
Brand	Final name, logo, phone number, opening hours, photos	Shop owner
Messaging provider	SMS or WhatsApp provider, cost, and WhatsApp business approval time	Dev
Rider capacity	Who the riders are and how fast they can be booked at peak	Shop owner
Legal	Terms of service, privacy policy, business registration for payment onboarding	Shop owner
Manual refunds	Process for refunds until API access exists	Shop owner
Scope creep	Scheduled orders and extra features stay out of v1	Dev

18. Milestones (one developer, rough estimates)

Milestone	Scope	Estimate
M1 Foundations	Monorepo, database, auth, CI, design tokens and base components	1 week
M2 Menu	Public menu pages, admin menu manager, image handling	1 week

Milestone	Scope	Estimate
M3 Cart and orders	Cart, checkout, quote, order creation, state machine	1.5 weeks
M4 Payments	Bachs sandbox test, checkout session, webhook, expiry, reconciliation	1.5 weeks
M5 Operations	Admin order board, rider booking, notifications, order tracking	1.5 weeks
M6 Extras	Promo codes, zones, store hours, pickup	1 week
M7 Launch	Testing, load check, security review, monitoring, soft launch	1 to 2 weeks

Total: about 8 to 9 weeks.